ITI 1120
Lab # 1

An Introduction to the Lab Environment

About TA (Teaching Assistant) …

- Name:
- E-mail:

Objectives

- Getting familiar with the lab environment
  - logging in and logging Out
  - email
  - Brightspace
  - Coursera
- Creating and Submitting a Practice Assignment 0 on Brightspace
- Enrolling into two Python courses on Coursera
- Writing your first Python program
- Two Python programming exercises (with Turtle graphics)
- Blockly exercises

Logging in and Logging out

LOG IN:

Follow the instructions on the screen to log in. In particular:

•Use your UofO account to authenticate. For example, if your student's email is jdoe123@uOttawa.ca, you will specify the username 'jdoe123' and use your UofO password.

LOG OUT:

•When you are all done and ready to leave, the last thing you **ALWAYS** do is to log out. Another student could have access to your files if you do not log out. Do not turn the power off after logout, leave the power on.

Accessing Brightspace

- What is at Brightspace?
  - All the course material is there (think of it as the course web page)
  - Downloading and submitting assignments
  - Announcements
  - Consulting your grades.
  - Discussion groups/forums
  - Students are required to check Brighhtspace course webpage and their OttawaU email frequently for announcements and all material.
- To access brightspace:
  - Go to https://uottawa.brightspace.com/d2l/home
  - Enter your user name and password and sign In.
  - Click on ITI1120 to access the course

Starting Lab 1

- Open a browser and log into Brightspace
- On the left hand side under Labs tab, find lab1 material contained in lab1-students.zip file
- Download that file to the Desktop and unzip it.

Starting Lab 1

- Open the file provided with this lab called OpeningIDLE.pdf

- Follow the 5 steps described in OpeningIDLE.pdf

In Python shell type:

- type 1+2*3, press enter and observe the output
- Then type (1+2)*3, press enter and observe the output
- Then type 2**10, press enter and observe the output
- Then type 11/2, press enter and observe the output
- Then type in the expression that sums first 5 positive integers
- Then type the expression that finds average of 25, 12, 40 and 1

Practice Submitting Assignment 0

- **First to learn how to create your submission follow the steps below:**
  - Open the file provided with this lab called UsingIDLE-first-program.pdf
  - Follow the 10 steps to create Python program A0_xxxxxxx. py and place it in the folder you crated called A0_xxxxxxx.
    (Note that what you see on your computers may differ slightly from that what you see in UsingIDLE-first-program.pdf)
  - Right-click on the A0_xxxxxxx folder and then select "Add to A0_xxxxxxx.zip".
    - This will create a zip file with the contents of the directory.

- **Then to submit (A0_xxxxxxx.zip) follow the steps below:**
  - Go to Brightspace uottawa.brightspace.com
  - On the left-hand side click on Assignments tab
  - Click on Assignment 0
  - Under "Submissions" you should see dashed rectangle labeled with "Drop files here, or click below!"
  - Drag and drop A0_xxxxxxx.zip there. You should now see A0_xxxxxxx.zip in that rectangle.
  - Scroll down and click on "Submit".
  - You will see a pop-up saying "+Submitted successfully". You will also receive a notification email.
  - You can submit more than once. Try it! The old submission will not be deleted.

Identification

- The following information must be included at the beginning of each program in your future assignments. For example, in Assignment 1, your program a1_xxxxxxx. py must start with:

```
# Course:  IT1 1120
# Assignment number
# Family name, Given name
# Student number
```

Finishing up with lab technicalities

- When you log out from a lab computers, the files and folders you created will disappear. Thus if you need them copy them to a USB memory stick before logging out.

- When you are ready to leave, double-click on the logout icon. Do not turn off the power.

- But, before logging out … complete the following tasks

Enroll into two Coursera courses

1. If you do not have Coursera account, first create one here and remember your login credentials: https://www.coursera.org/learn/learn-to-program?authMode=signup
2. Enroll into the following two free Python courses: (If there is some option to pay something – do not pay anything. Choose the free option.)

https://www.coursera.org/learn/learn-to-program

and

https://www.coursera.org/learn/program-code

Make sure you enroll now, as enrollment will not be possible after the posted enrollment deadline.

Turtle Program Example

- Start IDLE
- The material you downloaded for Lab 1 has a Python program in a file called `turtle_smiles.py`
- Open that file with IDLE
- Press Run Module and observe what happened.
- Now study the code and the following two slides before attempting to do Exercise 1 (explained in 3rd slides from here)
- Notice the lines in `turtle_smiles.py` that start with #. All lines starting with # will be ignored by python interpreter when your press run. Therefore I used # to explain parts of the program to you.

Useful functions in Turtle Class

Table CS.1 Some methods of the `Turtle` class. After importing the module `turtle`, you can obtain the full list of `Turtle` methods in your interactive shell using `help(turtle.Turtle)`

Usage	Explanation
`t.forward(distance)`	Move turtle in the direction the turtle is headed by `distance` pixels
`t.left(angle)`	Rotate turtle counterclockwise by `angle` degrees
`t.right(angle)`	Rotate turtle clockwise by `angle` degrees
`t.undo()`	Undo the previous move
`t.goto(x, y)`	Move turtle to coordinates defined by `x` and `y`; if pen is down, draw line
`t.setx(x)`	Set the turtle's first coordinate to `x`
`t.sety(y)`	Set the turtle's second coordinate to `y`
`t.setheading(angle)`	Set orientation of turtle to `angle`, given in degrees; Angle 0 means east, 90 is north, and so on
`t.circle(radius)`	Draw a circle with given `radius`; the center of the circle is `radius` pixels to the left of the turtle
`t.circle(radius, angle)`	Draw only the part the circle (see above) corresponding to `angle`
`t.dot(diameter, color)`	Draw a dot with given `diameter` and `color`
`t.penup()`	Pull pen up; no drawing when moving
`t.pendown()`	Put pen down; drawing when moving
`t.pensize(width)`	Set the pen line thickness to `width`
`t.pencolor(color)`	Set the pen color to color described by string `color`

This table is from "Introduction to Computing Using Python" by Lj. Perkovic

Useful functions in Turtle Class

Usage	Explanation
`s.bgcolor(color)`	Changes the background color of screen `s` to color described by string `color`
`s.clearscreen()`	Clears screen `s`
`s.turtles()`	Returns the list of all turtles in the screen `s`
`s.bye()`	Closes the screen `s` window

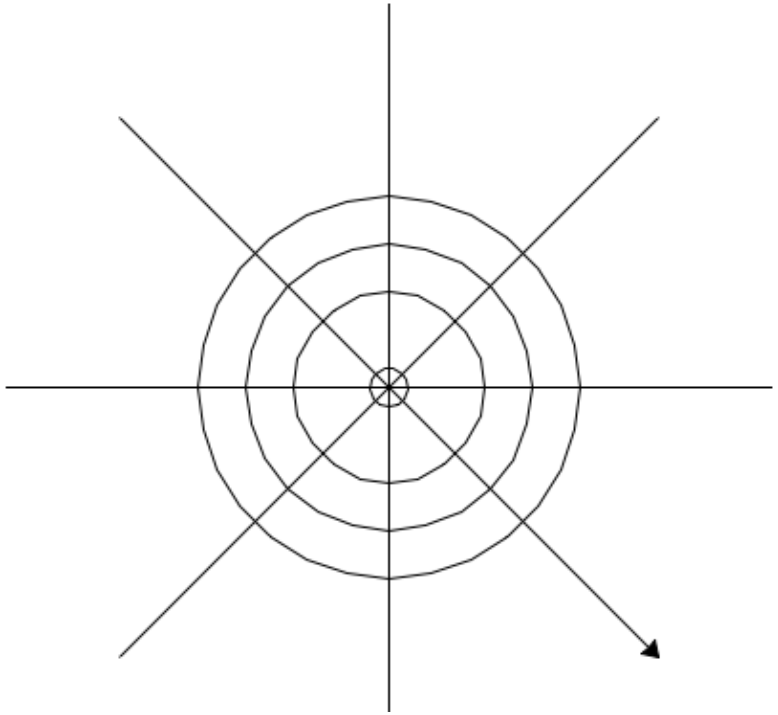
Table CS.2 Methods of the `Screen` class. Shown are only some of the `Screen` class methods. After importing module `turtle`, you can obtain the full list of `Screen` methods in your interactive shell using `help(turtle.Screen)`

This table is from "Introduction to Computing Using Python" by Lj. Perkovic

Exercise 1

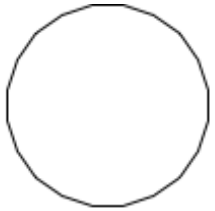
- From IDLE open file turtle-ex1.py that is provided with Lab 1.
- Add your own code inside of that file so that your final program draws a dart board similar to the one depicted below. (no need to use loops. copy/paste is your friend in this lab). Do not forget that whenever you make changes to a file, you have to save it first

  before you can run it.



Exercise 2

- From IDLE open file turtle-ex2.py that is provided with Lab 1.
- Add your own code inside of that file so that your final program draws an image similar to the one depicted below. (no need to use loops. copy/paste is your friend in this lab). Place your code where indicated in file turtle-ex2.py





Having fun:
write a program in Blockly

- What is Blockly: visual programming language developed by Google (to help students learn programming)

- Instructions are blocks that fit together (like lego bricks) to make up a computer program.

Final Exercises

•Use Blockly to solve level 9 and 10 mazes here:

https://blockly-games.appspot.com/maze?lang=en&level=9&skin=0

https://blockly-games.appspot.com/maze?lang=en&level=10&skin=0

-To get an idea on how to solve them (i.e. how to program in Blockly, you many need to solve a few earlier levels)

-Try to find solutions with the smallest possible number of blocks

-Challenge: Can you solve level 9 with 4 blocks. What about level 10 with 5!!